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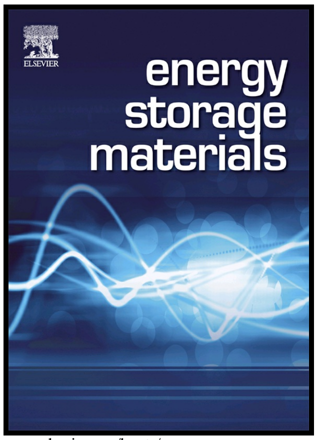

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3D Printed Separator for the Thermal Management of High-performance Li Metal Anodes

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Abstract

Lithium (Li) metal anodes with a high theoretical gravimetric capacity and a low redox potential have attracted considerable attention for Li-ion based batteries, including Li-sulfur (Li-S) and Li-oxygen (Li-O_2) batteries. The elimination of safety concerns and long cycling stability issues are the main challenges that must be solved to apply Li metal anodes in any application. Herein, we design a novel and safe separator for Li metal batteries by integrating thermally management boron nitride (BN) nanosheets into poly vinylidene fluoride-hexafluoropropene (PVDF-HFP) via an advanced extrusion based 3D printing technique. The BN-separator offers a uniform thermal distribution interface, which enables homogeneous Li nucleation, enhanced suppression of Li dendrite growth, and further improves the overall electrochemical performance. A stable Coulombic efficiency of 92% is maintained after 90 cycles at 1 mA cm^{-2} by using this novel separator in Li/Cu cells. In contrast, the Coulombic efficiency of the control cells decreases to 70% after 30 cycles at a current density of 1 mA cm^{-2}. In addition, in a Li/Li symmetric configuration, the cells using the 3D printed BN-separator exhibit improved cycling stability with lower voltage polarization over 500 h. Our findings on 3D printing for a thermally managing BN-separator can serve as a general strategy for further advanced rechargeable batteries.

Keywords: 3D Printing, Thermal Management Separator, Boron Nitride, Lithium Metal Anode

Introduction

With the development electric vehicles and energy storage stations, rechargeable batteries with high energy and power densities play a critical role in these energy storage appliactions.[1-5] Lithium (Li) metal anodes have attracted considerable attention for Li-ion based batteries in recent years, such as Li-S and Li-O_2 batteries, due to its high theoretical capacity (3860 mA h g^{-1}), and low redox potential (-3.04 V *vs* standard hydrogen electrode).[6-9] There are several intrinsic problems that hinder the widespread utilization of Li metal anodes in consumer markets.[10-14] During the Li plating/stripping process, the huge volume expansion and "hot-spots" can result in the formation of cracks in the solid electrolyte interface (SEI) layer, and can then further accelerate the electrode pulverization.[15-19] Moreover, the exposure of fresh Li metal to the electrolyte is usually accompanied by the irreversible side reactions, which reduces the Coulombic efficiency in long-term cycling.[20-23] In particular, the growth of Li dendrites and the generation of "dead" Li can lead to short circuits, and give rise to serious safety hazards, such as the generation of fumes, fire, and explosions.[24-28] To overcome these challenges, tremendous efforts have been dedicated to regulate Li nucleation and suppress Li dendrite growth [29], which includes exploring new electrolyte alternatives and additives [30-32], designing an interfacial protective layer on the surface of the Li metal [18], and the implementation of advanced separators.[33, 34] A functional separator in Li metal batteries can effectively suppress the growth of Li dendrites to improve the cycling performance and enhance the Coulombic efficiency.

Recently, advanced 3D printing technology, which can be used to design functional structures by combining computer-aided design and manufacturing procedures, has been regarded as a revolutionary fabrication process that encompasses laboratory to industry applications.[35-37] In comparison to traditional manufacturing methods, 3D printing possesses unique advantages in geometrical shape, as well as rapid prototyping, especially with complex 3D structure constructions.[38, 39] As a consequence, 3D printing is considered as an advanced manufacturing technique that is widely applied in many fields, but not limited to, electronics,[36] bioscience,[40] and energy storage [41]. Meanwhile, multiple studies also indicate that extrusion-based 3D printing is a facile and sophisticated strategy for materials design and fabrication that even extends to electrodes[42, 43] and rechargeable Li-ion micro-batteries.[44] Recently, Durstock and co-workers adopted 3D printing to design ceramic–polymer electrolytes for flexible Li ion batteries.[45] Thus, 3D printing can be implemented as a strategy to manufacture energy storage devices in a more practical way.

BN is regarded as one of the most important thermally conductive materials in energy storage devices, mainly owing to its thermal stability, electrically insulating properties, as well as its outstanding mechanical strength.[46-48] Herein, we report a thermal management separator by integrating BN with the 3D extrusion printing technique for Li metal batteries. This separator can also be extended to other battery chemistries, such as sodium-ion, Li-S and Li-O_2.[49-52] As shown in Figure 1a, BN nanosheets, once fully solvated in poly vinylidene fluoride-hexafluoropropene (PVDF-HFP), possess the optimal properties to print a separator (denoted as BN-separator), which upon printing creates a uniform thermal distribution interlayer consisting of a sprial pattern. The adjacent spiral lines become connected with each other to form an integrated film under the set spacing. To illustrate the printing process, the circle spiral model is chosen to print a circular separator with a radius of 1.25 cm; digital photos at different stages of the corresponding printing process are shown in Figure 1b. Separators with shapes such as a rectangle can also be printed under the facile procedure design (Figure S1 and S2). To further illustrate this process, a 6×10 cm film is also presented, as well as more details for printing process (Figure S3).

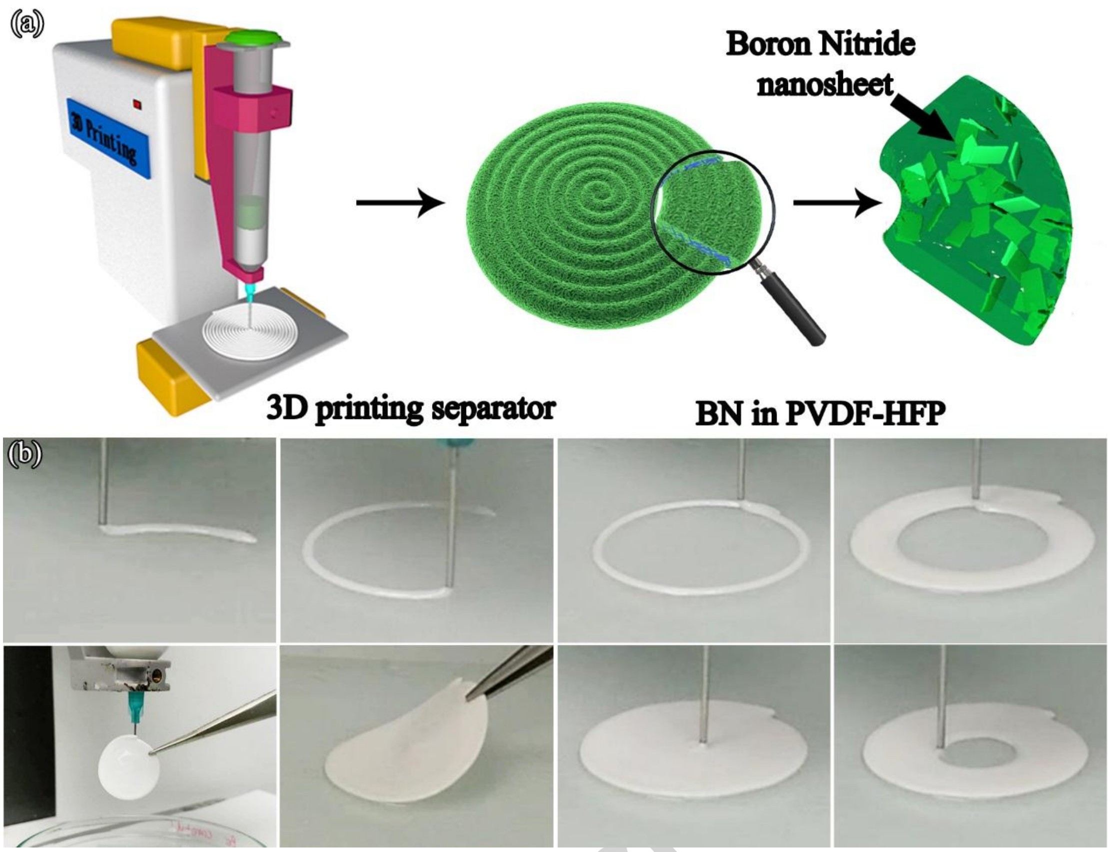

Figure 1. (a) Schematic illustration of the 3D printing apparatus, the BN in PVDF-HFP separator and the corresponding composition and structure; (b) digital photographs at different points in the printing process.

RESULTS AND DISCUSSION

The morphology and microstructure of the BN-separator is demonstrated in Figure 2. As presented in Figure 2a, the film with a helical and integrated structure can be observed in the low magnification field-emission scanning electronic microscopy (FE-SEM) image. The separator surface shows a porous network structure with porosity ranging from 1 to 5 μm, while the particle size is irregular and shows a range of sizes over several micrometers (Figure 2b-c). Furthermore, the cross-sectional morphology of the BN-separator (Figure 2d) displays a thickness of approximately 30 μm with the continuous porous structure. The separator can be easily permeated by the electrolyte, and the hierarchical pores inside form a continuous pathway for fast Li-ion transport between the anode and cathode (Figure 2e). Additionally, the thickness of the BN-separator is a critical factor for the shuttling of ions in a battery, which is directly related to overall battery performance. A thinner separator is beneficial to decrease the internal resistance and power consumption, although it will lose the adequate mechanical strength and increase the risk of short-circuit in the cells. In our case, the thickness of the separator can be easily controlled through the advanced 3D printing technology, by adjusting the air-powered fluid dispenser under a specific printing rate. When the air-powered fluid dispenser was adjusted to 5 Pa at a rate of 0.7 mm/min, a BN-separator with a thickness of 29 μm can be obtained (tested by vernier calipers, Figure S3). This measured thickness directly agrees with the FE-SEM thickness observations, and can also meet the requirements of commercial separators. As a comparison, the thickness of a commercial separator is provided in Figure S3, which is about 20 μm. After changing the printing pressure to 7 and 10 Pa, BN-separators with thicknesses of 50 μm and 100 μm can be achieved, respectively, as shown in Figure S3. The pore structure of the BN-separator is an important factor for battery performance. The overall porosity is approximately 73% according to the gravimetric method, and the pore size is about 3.9 nm. The thermal stability of the BN-separator has been characterized via a thermal-shrinkage test, as shown in Figure S4. At a temperature of 110 ^{o}C, the separator almost maintains the original size without shrinkage, while the shrinkage is 6% when treated at 150 ^{o}C for 0.5 h.

The as-printed BN-separator is dried in a vacuum oven for 12 h, and is then punched out into a disk with a radius of 8 mm, and is then used in a coin cell. The as-prepared BN-separator can be easily bent and subsequently recovered to the original shape (Figure 2f-2g), indicating good flexibility. Furthermore, the electrolyte penetration property of the BN-separator and commercial separator are compared by single-droplet experiment using 1 M $LiPF_6$ in ethylene carbonate (EC) and diethyl carbonate (DEC). As shown in Figure 2h, the droplet stays in the center of control separator without spreading out or soaking in completely. In sharp contrast, the BN-separator absorbs the droplet of electrolyte quickly, indicating the improved penetration property in the electrolyte (Figure 2i).

The structures are characterized by X-ray diffraction (XRD) measurements. The XRD patterns of the commercial BN, PVDF-HFP, and BN-separator are shown in Figure 2j and Figure S5, respectively. The diffraction peaks at 2θ values of 13^o, 27^o, 42^o, 50^o and 55^o in Figure 2j can be ascribed to the phase structure of BN, compared with the original samples in Figure S5. Moreover, the diffraction peak at ~20^o is assigned to the PVDF-HFP (Figure S5), illustrating the existence of PVDF-HFP in BN-separator. In addition, the intensity and half-width at half-maximum at 13^o exhibits a prominent decrease, indicating the homogeneous mixing of BN with PVDF-HFP in the BN-separator, which agrees well with the FE-SEM results. The Fourier transform infrared spectra (FTIR) spectra of the commercial BN, PVDF-HFP, and BN-separator are also characterized, as shown in Figure 2k and Figure S5. The characteristic bands in the range from 800 to 1500 cm^{-1} in Figure 2k further confirms the presence of PVDF-HFP in the 3D printed separator. An evident absorption peak at around 770 cm^{-1} is ascribed to the B–N bonds with B–N stretching vibration and B–N–B bending vibration.[53] Compared to the peak at ~750 cm^{-1} for pure BN power (Figure S5), the absorption peak shifting towards the longer wavelength side indicates the existence of BN in BN-separator which is in accordance with the above XRD analysis.

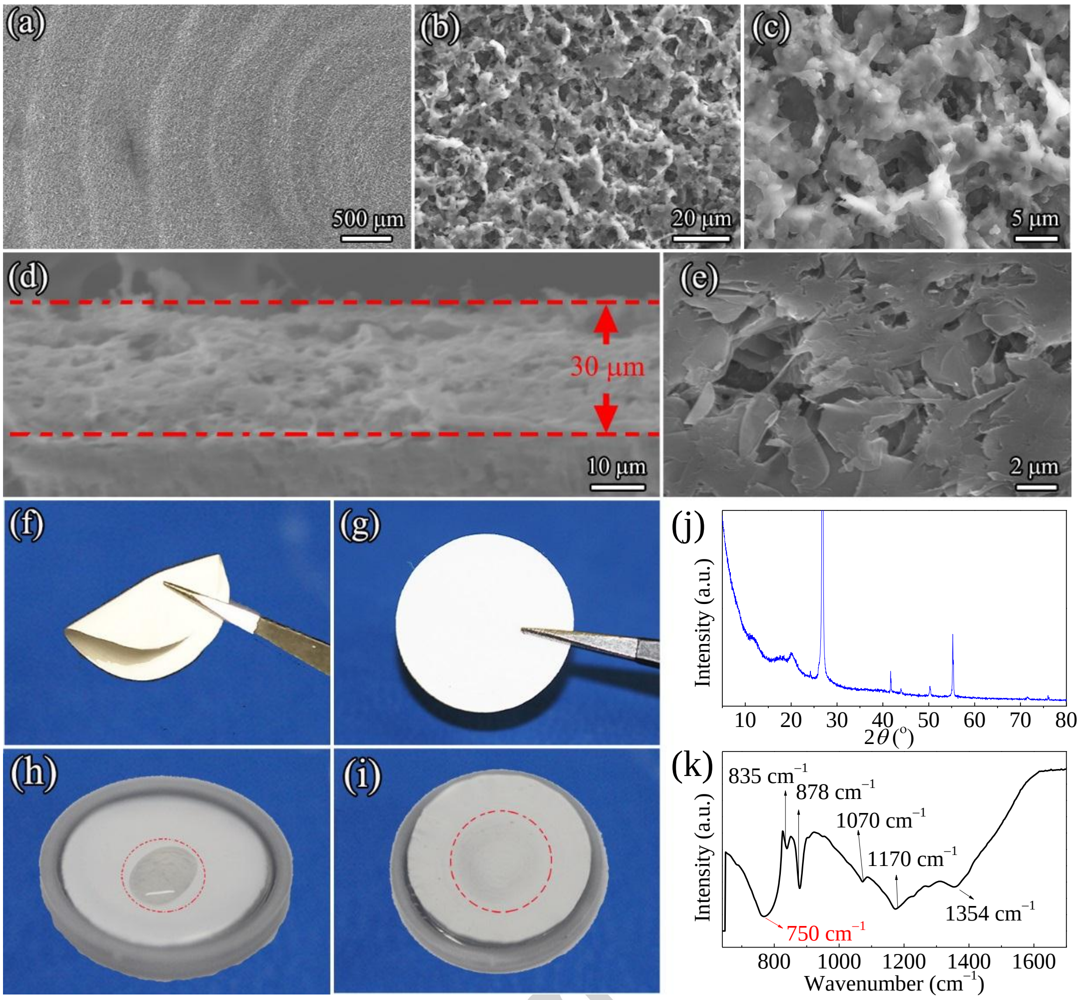

Figure 2. SEM images for the (a-c) top surface and (d, e) cross-section of the BN-separator; digital photographs of (f) a twisted and (g) recuperated BN-separator; electrolyte wettability behaviors of (h) control-separator and (i) BN-separator; (j) XRD pattern and (k) FTIR spectrum of BN-separator.

To study the electrochemical performance of the BN-separator, coin cells (BN cells) were assembled, using copper foil |BN-separator| Li foil as the working electrode, separator and counter electrode, respectively. The control cells refer to the cells using control separator. Coulombic efficiency is an important factor to indicate and evaluate the application possibility of the separator. Figure 3a presents the Coulombic efficiencies of BN-separator and control cells at 0.5 mA cm^{-2} with a plating capacity of 0.5 mA h cm^{-2}. The BN-separator cell shows good cycling performance upon 10 hours with high Coulombic efficiencies of ~ 90% after ten cycles, which remains at about 92% for over 90 cycles. On the other hand, the Coulombic efficiency of the control cell increases to 92% in the 2nd cycle, and then drops below 80% after only 30 cycles. When the current density increases to 1 mA cm^{-2} with a plating capacity of 1 mA h cm^{-2}, the BN cell can achieve a high Coulombic efficiency of 90% after 2nd cycle, which finally stabilizes at around 92% after 90 cycles, while the control cell shows a Coulombic efficiency of about 87% at the second cycle, and decreases to 70% after about 30 cycles (Figure 3b). The electrochemical results demonstrate that the cell with the BN-separator has better cycling performance and higher Coulombic efficiency than that of the commercial separator in Li metal batteries. Detailed plating/stripping behaviors in the BN and control cells were investigated, which are shown in the charge and discharge profiles in Figure 3c and 3d. After 90 cycles, the control cell shows a craggy profile, indicating a formation of Li dendrites. The BN-separator is facile to regulate Li nucleation, suppress Li dendrite growth, and enhance the electrochemical performance (Figure S6). These results demonstrate that the BN-separator can decrease the risk of short-circuiting and provide stable electrochemical performance.

Li/$LiFePO_4$ cells are fabricated to investigate the electrochemical properties with BN-separator and control separator. Figure 3e and 3f show the first three charge and discharge curves of the Li/$LiFePO_4$ cells using BN-separator and control separator, respectively. The current density is 0.2 C (1 C=170 mA h g^{-1}) and the cut-off voltage range is from 2.5 to 4.2 V. A flat voltage plateau is evident around 3.4 V, which corresponds to the typical Fe^{2+}/Fe^{3+} redox reaction accompanied with the insertion and extraction of Li ions. The specific capacities of the cell with BN-separator at 0.2 C are 139.4, 145.0, and 147.3 mA h g^{-1} in the first three cycles, which is a little higher than the commercial separator. This result further suggests that BN-separator displays better electrochemical properties than that of the commercial separator.

The electrochemical stability of BN-separator is further evaluated in Li/Li symmetric cells. Figure 3g presents the time-dependent voltage profiles for Li/Li symmetric cells with the BN-separator and control separator under a constant current density of 1 mA cm^{-2}. As depicted, the cell with the BN-separator delivers a more stable voltage curve with a small hysteresis of ~150 mV for up to 500 h, compared with the cells using the commercial separator (~300 mV). This result indicates that the formation of Li dendrites is suppressed during the long cycling time. The detailed voltage profiles are presented in Figure 3h-3j with a time period of three hours at different stages, respectively. For the control separator, the lithium stripping/plating overpotential is approximately 550 mV for the initial discharge process, which is higher than that of BN-separator (~150 mV), as shown in Figure 3h. During the second and third cycle, a sharp peak and a relatively small plateau (marked with arrows) can be seen in Figure 3h for the control separator, which could be attributed to Li nucleation in unsystematic locations across the surface of Li metal electrode.[54] Three hundred hours later, the arc-like profiles evolve with the Li plating/stripping to form the SEI film and dead Li layer.[55] Moreover, the overpotential in control cells is about 120 mV, which is higher than that in BN-separator (75 mV) at 300 hours, as shown in Figure 3i. Even after 500 h, the cell with BN-separator presents extremely stable profiles and lower voltage polarization profile than the commercial separator under the same current density of 1 mA cm^{-2}. These results further indicate that BN-separator in the cell is beneficial to restrain the growth of Li dendrites during a long cycling process.

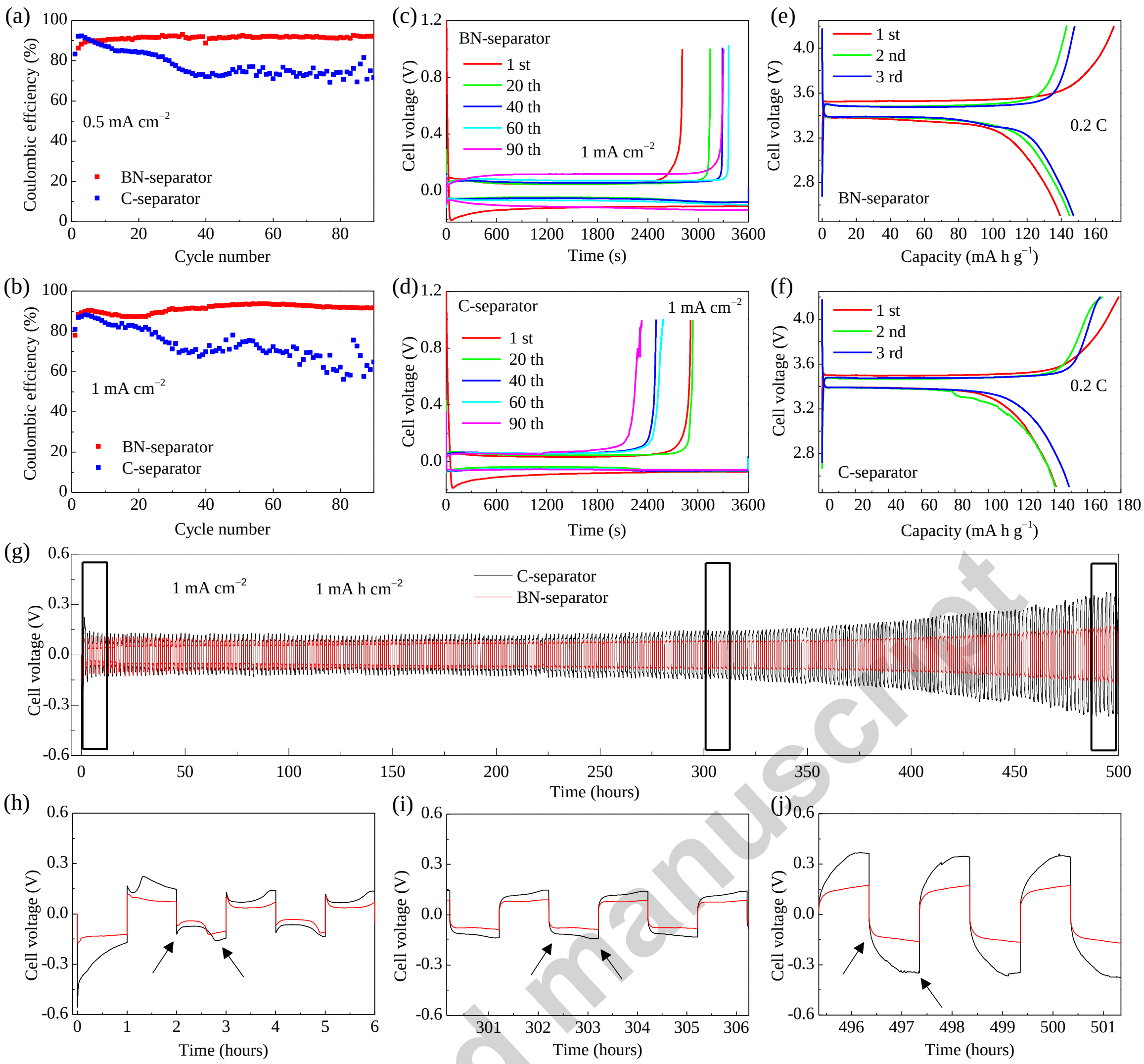

Figure 3. Electrochemical performance the BN-separator and control separator: Coulombic efficiencies of the Li/Cu cells at current densities of (a) 0.5 mA cm^{-2} and (b) 1 mA cm^{-2}, (c, d) the corresponding voltage profiles of different cycles at 1 mA cm^{-2}; (e, f) voltage profiles of Li/$LiFePO_4$ cells for the first third cycles; (g-j) voltage profiles of the Li/Li symmetric cells at a current density of 1 mA cm^{-2}.

The thermal management performance of the BN-separator is analyzed to disclose the function of BN in improving the cycling stability in Li metal batteries. Figure 4a shows a schematic of the experimental setup, which involves a laser power input as heat source, a substrate to hold separators, and a FLIR Merlin MID infrared (IR) camera to detect the maximum temperature of the separators when exposed to laser heat source. The IR camera has a resolution of 320 x 256 pixels, which is controlled by ThermCAM Research software. To ensure the same input energy and accurate temperature readings from IR camera for the comparative samples, a graphite layer is coated at separator center, which is slightly larger than the laser spot size (about 1 mm). The temperature distribution profile and the max temperature at the center are presented in Figure 4b-g, respectively. The control sample shows a brighter hotspot color compared to that of the BN-separator, indicating the heat is accumulated at the separator center due to its poor thermal dissipation ability. Different laser powers were tested for both samples for validation purposes. When the laser source is incident with a power of 0.073 W, the maximum temperature of BN-separator and control separator are 80.3, and 86.8 ^{o}C, respectively (Figure 4f), which shows 6.5 ^{o}C temperature difference between the two separators. The temperature difference is further increased to 14 ^{o}C with an incident laser power of 0.093 W, which validates the enhanced thermal management performance of the BN-separator when exposed to a heat source. As shown in Figure 4h, when the local heat source is incident on the separator, the localized heat can be diffused effectively along the BN-separator yielding a lower central temperature. In contrast, the control separator presents a higher central temperature due to its insufficient thermal transport properties. Obviously, the superior thermal management performance of the BN-separator is desirable to alleviate the heat generation issue during the charge and discharge process. Compared with the control separator, the BN-separator provides a more homogenous temperature distribution to enable uniform nucleation and growth of Li, and thus leads to better cycling stability and higher Coulombic efficiency. Moreover, the uniform thermal distribution of the BN-separator can curb the risk of short circuit during the charge and discharge process, thus decreasing the corresponding risk of fire.

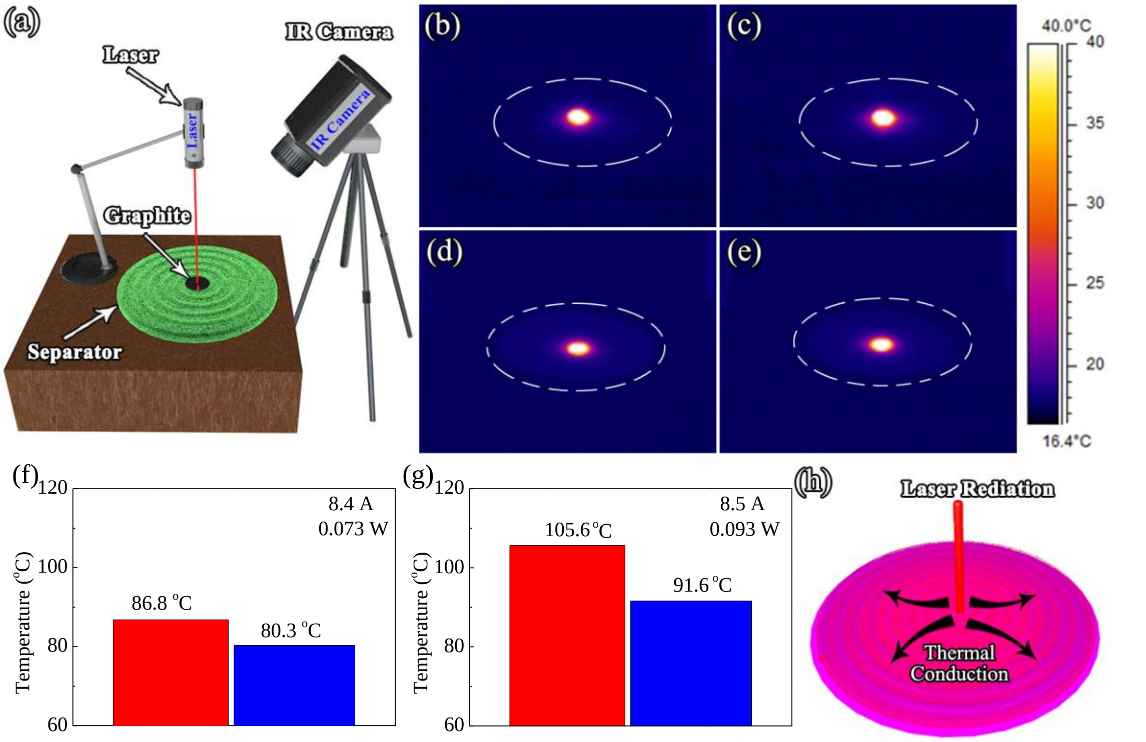

Figure 4. (a) Illustration of the thermal management experimental setup on the separator, with the corresponding IR temperature distribution mapping images of (b, c) control separator and (d, e) BN-separator at laser source power of 0.073 W and 0.093 W, respectively, and (f, g) the corresponding maximum temperatures on the BN-separator and control separator from the IR camera, and (h) schematic of heat dissipation from center of BN-separator when exposed to local heat source.

CONCLUSIONS

In summary, we designed a thermal management separator for Li metal batteries using an extrusion-based 3D printing technology. The BN-separator can provide fast heat dispersion and a uniform thermal distribution at interface during the Li stripping/plating process, to enhance the electrochemical performance. Compared with the control separator, the BN-separator in Li/Cu batteries can provide a very stable cycling performance and high Coulombic efficiency. In Li/$LiFePO_4$ cells, the 3D printing BN-separator demonstrates stable electrochemical properties, but also maintains a high specific capacity compared with the commercial separator. Furthermore, the Li symmetric cells with the BN-separator presents more stable voltage profiles with lower overpotential than the control separator for over 500 hours of cycling. This work provides an effective manufacturing strategy to fabricate thermally managing separators for Li metal batteries.

EXPERIMENTAL SECTION

Methods

Preparation of PVDF-HFP/boron-nitride (BN) Composite inks: 1.5 g BN was dispersed in 4.5 g dimethyl formamide (DMF) in a bottle. Then, the as-obtained BN suspension solution is directly dispersed in an ultrasonicator for 12 h. After that, 1 g PVDF-HFP was added into the suspension solution under stirring. The mixed suspension was heated to 60 ^{o}C and stirred for 12 h. Finally, the PVDF-HFP/boron-nitride (BN) composite ink was obtained for further use.

The 3D Printing of PVDF-HFP/boron-nitride (BN) separator: The preparation of PVDF-HFP/boron-nitride (BN) Composite ink was transferred into a 30 mL syringe. The syringe was then mounted in a 3D printer (Fisnar F4200N Robot) and the printing rate and form can be adjusted by a setting procedure. The thickness of the separators also can be controlled by an air-powered fluid dispenser (Fisnar DSP501N) and the printing rate. After printing, the separators were dried directly under room temperature. Finally, the separators were cut into circular membrane with a diameter of 16 mm to assemble coin cells.

Material and electrochemical characterizations: The morphology of the samples was performed on Hitachi SU-70 field emission scanning electron microscopy and the elemental composition was detected on Inca X-Max 50 (Oxford Instrument) equipment with an energy-dispersive spectrometer (EDS). XRD patterns were conducted by the D8 Advanced (Bruker AXS, WI, USA). Fourier transform infrared (FTIR) spectrometry were measured on Bruker VERTEX 70 equipment.

The porosity (ε, %) of the BN-separator was determined by the weight of absorbed water in the pores using gravimetric method.[56]

$$\varepsilon = (\Delta m/\rho)/V \quad (1)$$

where Δm is the mass difference between the wet membrane and the mass of the absorbed water, ρ is the density of water and V is the volume of dry separator.

The mean pore radius (r_m) of the separator is determined by the Guerout–Elford–Ferry equation, using the pure water solution.

$$r_m=\sqrt{\frac{(2.9-1.75\varepsilon)8\eta lQ}{\varepsilon\times A\times\Delta P}} \tag{2}$$

where η is the viscosity of water under 25 ^{o}C, Q is the volume of infiltrated water per unit time (m^3/h), A is the effective area of the separator (m^2), and ΔP is the operational pressure (50 cm Hg).

The anti-thermal-shrinkage performance of the BN-separator was determined by testing the dimensional variation at various temperatures for 0.5 h, and calculated according to the following equation:

$$\text{Shrinkage}(\%)=(S_0-S_T)/S_0\times100\% \tag{3}$$

where S_0 and S_T are the original separator area and the separator area after thermal treatment at various temperatures, respectively.[56]

The CR 2032-type coin cells were assembled using 1.0 mol L^{-1} $LiPF_6$ in a mixed solvent of ethylene carbonate and diethylene carbonate (v/v = 1:1) as electrolyte. The galvanostatic cycling performances were performed on Arbin BT2000 system.

ASSOCIATED CONTENT

Supporting Information

ACKNOWLEDGMENTS

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